

SDE 1 | Machine Learning Engineer | Data Scientist | Business Analyst | AI Enthusiast

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Professional Summary:

Machine Learning Engineer with 3+ years developing production-ready AI solutions processing 100K+ daily transactions, adhering to rigorous software development life cycle and code reviews. Proven track record delivering 95% fraud detection accuracy and 60% faster deployment times via MLOps automation.. Published researcher with expertise in TensorFlow, AWS, and Apache Spark, seeking to drive data-driven business growth through scalable ML infrastructure

Technical Skills

Programming: Java, C/C++, Python, R, JavaScript, SQL

Libraries/Frameworks: Scikit-learn, TensorFlow, XGBoost, PyTorch, Pandas, NumPy, Apache Spark, Apache Airflow, Flask, Streamlit, FastSpeech 2, HiFi-GAN, LSTM, Matplotlib, Seaborn, ONNX, TensorRT, Outlier Detection

Database: MongoDB, PostgreSQL, Redis, SQLite

Developer Tools: Git/GitHub, Visual Studio Code, Snowflake, Linux, Docker, Jupyter.

PROJECTS

E-Commerce clickstream analysis - Data Scientist/Machine Learning

Jan 2025 - Apr 2025

- Designed feature pipelines processing 50K+ customer records using pandas and NumPy for RFM analysis.
- Implemented ensemble methods (Random Forest, XGBoost) and deep learning (LSTM, Dense Neural Networks) employing TensorFlow/Keras, rigorously testing models using algorithm design and data structures best practices, achieving 85%+ F1-score by GridSearchCV hyperparameter tuning and 5-fold cross-validation.

AI-Based Natural Speech Synthesis from Text – AI/ML

Dec 2024 - Jan 2025

- Developed FastSpeech 2 encoder-decoder with multi-head attention, duration predictor and HiFi-GAN vocoder using PyTorch, employing object-oriented design principles for a robust distributed systems architecture, delivering mel-spectrogram generation and neural vocoding algorithms.
- Accelerated model inference with ONNX quantization, TensorRT GPU optimization achieving <200ms latency, deployed through Flask REST API featuring asyncio WebSocket streaming and Docker containerization.

Stock Market Trend Prediction – Machine Learning

Jan 2023 - May 2023

- Spearheaded development of a comprehensive fraud detection system by engineering multi-layered risk indicators including velocity checks and behavioral anomaly detection, reducing false positives by 30% through machine learning optimization..
- Launched real-time prediction dashboard utilizing Flask and AWS EC2 featuring Nginx reverse proxy, demonstrating scalable cloud deployment and business intelligence capabilities.
- Orchestrated real-time ETL pipeline using Apache Airflow DAGs, yfinance API integration, **deployed** on AWS EC2 with Nginx reverse proxy and PostgreSQL database for time-series storage.

PROFESSIONAL WORK EXPERIENCE

Celonis Process Mining - Virtual Intern

Dec 2022 – Apr 2023

- Reduced operational risk by 35% through Celonis platform analytics, identifying \$50k in potential fraud losses across business workflows.
- Accelerated C-suite decision-making by 40% leveraging executive dashboards featuring real-time KPI visibility.

AWS AI/ML Engineer - Virtual Intern

Feb 2022 – May 2022

- Constructed 3 end-to-end fraud detection pipelines leveraging AWS, AI, and ML technologies (SageMaker and TensorFlow), processing 100K+ daily transaction records, achieving 95% fraud classification accuracy and reducing investigation time by 60%.
- 3 enterprise-grade ML systems handling 50K+ daily requests featuring 99.2% uptime.

AWS Cloud Engineer - Virtual Intern

Aug 2021 – Dec 2021

- Architected cloud infrastructure deployment supporting 5K+ concurrent users by implementing automated EC2, S3, and VPC provisioning through CloudFormation IaC, achieving 60% faster deployment cycles and reducing operational overhead.
- Engineered CI/CD pipelines employing CodePipeline, CodeBuild, and Jenkins, leveraging robust source control management and build processes, automating software releases and accelerating delivery cycles by 40%, enabling faster time-to-market.

PUBLICATIONS & ACHIEVEMENTS

- Stock Market Trend Prediction Using Machine Learning** - IJCRT Journal Publication (Machine Learning & Financial Forecasting) ([IJCRT2304690](#))
- Customer Segmentation Using Machine Learning** - IJSREM Journal Publication (Clustering Algorithms & Business Intelligence) ([IJSREM17420](#))
- Bronze Badge** - CodeChef Platform - Solved 100+ DSA problems demonstrating strong algorithmic skills.
- Autonomous Navigation and Obstacle Detection System** - National Conference on Advanced Technologies in Computer Science (NCATCS), ROLLSTUHL 6.

EDUCATION

University of Texas at Arlington
MS in Data Science | CGPA: 3.66
Sreenidhi Institute of Science and Technology
B. Tech in Computer Science | CGPA: 8.34

Aug 2023-May 2025
Arlington, Texas
Aug 2019 – Apr 2023
Hyderabad, Telangana